STACKS, MODULI OF CURVES: LEMMA-CM-1-CURVES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1007-1029 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 1031-1057, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_moduli-curves.json (pb=1013, stmt 1007-1029, proof 1031-1057).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0E0I: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 967–976. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E0E: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 677–688. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E11: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1057–1057. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06LR: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1057–1057. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NS: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 1057–1057. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *There exist an open substack $\mathcal{C}urves^{CM,1} \subset \mathcal{C}urves$ such that*

(1) *given a family of curves $X \to S$ the following are equivalent*
   (a) *the classifying morphism $S \to \mathcal{C}urves$ factors through $\mathcal{C}urves^{CM,1}$,*

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(b) *the morphism $X \to S$ is Cohen-Macaulay and has relative dimension 1 (Morphisms of Spaces, Definition 06LR),*

(2) *given a scheme X proper over a field k with $\dim(X) \leq 1$ the following are equivalent*
   (a) *the classifying morphism $\mathrm{Spec}(k) \to \mathcal{C}urves$ factors through $\mathcal{C}urves^{CM,1}$,*
   (b) *X is Cohen-Macaulay and X is equidimensional of dimension 1.*

Proof. By Lemma 0E0I it is clear that we have $\mathcal{C}urves^{CM,1} \subset \mathcal{C}urves^{CM}$ if it exists. Let $f : X \to S$ be a family of curves such that f is a Cohen-Macaulay morphism. By More on Morphisms of Spaces, Lemma 0E11 we have a decomposition

$$X = X_0 \amalg X_1$$

by open and closed subspaces such that $X_0 \to S$ has relative dimension 0 and $X_1 \to S$ has relative dimension 1. Since f is proper the subset

$$S' = S \setminus f(|X_0|)$$

of S is open and $X \times_S S' \to S'$ is Cohen-Macaulay and has relative dimension 1. Moreover, formation of S' commutes with arbitrary base change because this is true for the decomposition above (as relative dimension behaves well with respect to base change, see Morphisms of Spaces, Lemma 04NS). Thus we get the open substack with the desired properties by the method discussed in Section 0E0E. □